DEVELOPING SPARSE RAMSEY THEORY

MASTER INTERNSHIP PROPOSAL

1. Internship Environment

The internship will be supervised by Ararat Harutyunyan and Lucas Picasarri-Arrieta, and will be held at LAMSADE (Université Paris-Dauphine), in the "pôle Optimisation, Structure et Algorithmes". The research can be conducted in either English or French. Feel free to contact both of us for any further information.

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2. Candidate Profile

The ideal candidate should be an excellent Master's student with a strong interest in structural graph theory. The project is deliberately open-ended, with several possible directions to explore depending on the candidate's interests. Its breadth also makes it a promising starting point for further research, potentially at the PhD level.

3. Introduction

In 1930, Ramsey [8] obtained the following seminal result. The *complete graph* on t vertices is denoted by K_t.

Theorem 1 (Ramsey [8])**.** *There exists a function $f\colon \mathbb{N} \to \mathbb{N}$ such that the following holds. For every integer $t \in \mathbb{N}$, every 2-edge-colouring of $K_{f(t)}$ yields a monochromatic copy of K_t.*

Ramsey's theorem has found applications in numerous mathematical fields, including number theory, ergodic theory, geometry, information theory, algorithms, complexity, and logic, see for instance [9] for a survey. This made it one of the most important tools in combinatorics and initiated the field now known as *Ramsey theory*, which seeks regularity amid disorder.

There exist numerous generalisations, variations, or analogues of Ramsey's theorem to many other combinatorial objects, which include for instance words [5], hypergraphs [7], tournaments [4], or ordered graphs [1]. This research project focuses on generalisations of Ramsey's theorem to edge-colourings of general graphs, instead of complete graphs only. A first step in this direction follows from the seminal result obtained by Kóvari, Sós, and Turán [6] in 1954. The *complete bipartite graph* with parts of size a and b is denoted by $K_{a,b}$.

Theorem 2 (Kóvari, Sós, and Turán [6])**.** *There exists $f\colon \mathbb{N} \to \mathbb{N}$ such that the following holds. For every integer t and every graph G, if*

$$|E(G)| \geqslant f(t) \cdot n^{2-\frac{1}{t}},$$

then G contains a copy of $K_{t,t}$.

It follows from the Kóvari-Sós-Turán Theorem that Ramsey's theorem can be generalised to edge-colourings of *dense graphs*, that is, graphs whose number of edges is quadratic in their order. For a real number $\varepsilon > 0$, we say that a graph G of order n is *ε-dense* if $|E(G)| \geqslant \varepsilon n^2$.

Corollary 3. *For every $\varepsilon > 0$, there exists $f_\varepsilon \colon \mathbb{N} \to \mathbb{N}$ such that the following holds. Let $t \in \mathbb{N}$ and G be an ε-dense graph of order $n \geqslant f_\varepsilon(t)$. Every 2-edge-colouring of G yields an induced monochromatic copy of K_t or $K_{t,t}$.*

Note that Corollary 3 indeed implies Ramsey's theorem, since complete graphs on at least three vertices are $\frac{1}{3}$-dense, and complete graphs do not contain any induced copy of $K_{t,t}$.

While the case of dense graphs is thus well-understood, the case of *sparse graphs*, that is, graphs with roughly linearly many edges, remains widely unexplored. In this direction, Char, Kawarabayashi, and Picasarri-Arrieta [3] recently obtained the following result. We denote by $\delta(G)$ the *minimum degree* of G.

Theorem 4 (Char, Kawarabayashi, and Picasarri-Arrieta [3])**.** *There exists $f \colon \mathbb{N} \to \mathbb{N}$ such that the following holds. Let $t \in \mathbb{N}$ and G be a graph with $\delta(G) \geqslant f(t)$. Every 2-edge-colouring of G yields an induced monochromatic subgraph H of G with $\delta(H) \geqslant t$.*

Observe that Theorem 4 again generalises Ramsey's theorem, as a complete graph is large if and only if it has large minimum degree, and any induced subgraph of a complete graph is complete. Theorem 4 appears as a first milestone to a potentially fruitful line of research, and the general purpose of this research project is to explore it. We conclude this document with a few specific questions.

4. Tightening and Strengthening Theorem 4

The proof of Theorem 4 is mostly existential, and does not make any effort in minimising the obtained function f. It yields that $f(t)$ can be taken to be at most triply exponential in t, while the classical random construction for Ramsey's theorem only yields a single-exponential lower bound in t. It would be interesting to capture the actual behaviour of the optimal function f.

Problem 5. Close the gap $2^{\Omega(t)} \leqslant f(t) \leqslant 2^{2^{2^{O(t)}}}$ for the smallest f satisfying Theorem 4.

It is also possible that a more precise version of Theorem 4 can be obtained, providing some information on the order of induced subgraph H.

Problem 6 (Suggested by Chepoi)**.** Prove (or disprove) the existence of $f \colon \mathbb{N} \to \mathbb{N}$ for which the following holds. Let $n, t \in \mathbb{N}$ and G be a graph with $|V(G)| \geqslant f(n)$ and $\delta(G) \geqslant f(t)$. Every 2-edge-colouring of G yields an induced monochromatic subgraph H with $|V(H)| \geqslant n$ and $\delta(H) \geqslant t$.

Also, it is not clear whether Theorem 4 extends to hypergraphs.

Problem 7. Prove (or disprove) the existence of $f \colon \mathbb{N}^2 \to \mathbb{N}$ for which the following holds. Let $r, t \in \mathbb{N}$ and G be an r-uniform hypergraph with $\delta(G) \geqslant f(r,t)$. Every 2-edge-colouring of G yields an induced monochromatic subhypergraph H with $\delta(H) \geqslant t$.

5. Ramsey-suitable graph parameters

Let γ be any graph parameter which is unbounded on complete graphs. We say that γ is *Ramsey-suitable* if there exists a function $f \colon \mathbb{N} \to \mathbb{N}$ such that every 2-edge-colouring of a graph G with $\gamma(G) \geqslant f(t)$ yields an induced monochromatic subgraph H of G with $\gamma(H) \geqslant t$.

Since we only consider graph parameters that are unbounded on complete graphs, any result of the form "*Parameter γ is Ramsey-suitable*" indeed generalises Ramsey's theorem. The *biclique number* $\tau(G)$ of a graph G is the largest integer t such that $K_{t,t}$ is a (not necessarily induced) subgraph of G. Ramsey's theorem, Corollary 3, and Theorem 4 can be reformulated in terms of suitability as follows.

Theorem 8 (Ramsey [8])**.** *The order of a graph is Ramsey-suitable.*

Corollary 9 (Kóvari, Sós, and Turán [6])**.** *The biclique number of a graph is Ramsey-suitable.*

Theorem 10 (Char, Kawarabayashi, and Picasarri-Arrieta [3])**.** *The minimum degree of a graph is Ramsey-suitable.*

Clearly, not all graph parameters are Ramsey-suitable. For instance, the *treewidth* is not Ramsey-suitable, as the $(k \times k)$-grid can be 2-edge-coloured in such a way that every induced monochromatic subgraph is a path. A natural candidate for Ramsey-suitability is the *chromatic number*, but a recent construction of Carbonero, Hompe, Moore, and Spirkl [2] rules it out.

Theorem 11 (Carbonero et al. [2])**.** *For every $k \in \mathbb{N}$, there exists a graph G with $\chi(G) \geqslant k$ and an edge-colouring of G for which every induced monochromatic subgraph of G is 4-colourable.*

It would be interesting to understand what graph parameters are indeed Ramsey-suitable.

Problem 12. Is there a characterisation of Ramsey-suitable graph parameters?

The first graph parameters to look at would be the *fractional chromatic number*.

Problem 13. Is the fractional chromatic number Ramsey-suitable?

6. Almost Ramsey-suitable graph parameters

While there exist both Ramsey-suitable and non Ramsey-suitable graph parameters, there might exist a category lying somewhere in between.

We say that a graph parameter γ is *almost Ramsey-suitable* if, for any real number $\varepsilon > 0$, there exists $f_\varepsilon \colon \mathbb{N} \to \mathbb{N}$ such that the following holds. For every $t \in \mathbb{N}$, if G is a graph with $\gamma(G) \geqslant f_\varepsilon(t)$ and G is 2-edge-coloured uniformly at random, then

$$\mathbb{P}\big(\exists H \subseteq_{\text{ind}} G, H \text{ monochromatic}, \gamma(H) \geqslant t\big) \geqslant 1 - \varepsilon.$$

Observe that Ramsey-suitable graph parameters are indeed almost Ramsey-suitable. While χ is not Ramsey-suitable, it is open whether it lies in this third category.

Problem 14. Is the chromatic number almost Ramsey-suitable?

This is related to a long-standing open question due to Bukh (Problem 2 here) suggesting that a graph G of chromatic number χ which is edge-coloured uniformly at random contains a (non necessarily induced) subgraph with chromatic number $\chi/\log\chi$ with high probability.

References

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